

PRODUCT SPECIFICATION

Battery Module / Pack

For (Power Bank / Home ESS)

Product	LFP Module / Pack
Model	RPM-5120V314AS1
Spec.	51.2 V / 314 Ah

Prepared	QA Checked	Approved
/	/	/

Customer Approval	Name	Signature	Date
	Company Name / Stamp		

Remplir Inc.

Revision History

Ver.	Date	Mark	Revision Description	Originator
V1.0	2026-06-01	/	First Edition	QA/QC Edward

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PART 1.
Scope

This specification defines and applies to the design specifications for 1P16S home ESS battery packs and inverters using 314Ah LFP battery cells supplied by Rimplier, Inc. These battery packs and inverters are manufactured for the purpose of electrical specifications and product supply, and do not include reliability testing such as acceleration or harsh conditions. If the customer requires special product specifications beyond this, they must proceed through a separate agreement with Rimplier, Inc.

Model name : RPM-5120V314AS1

PART 2.
Definition of Terms

No.	Terms	Definitions
2.1	Product	In this specification, "Product" refers to a (51.2 V - 314 Ah) / LFP battery pack manufactured by REMPLIR.INC (hereinafter REMPLIR).
2.2	Ambient Temperature	It refers to the temperature of the surrounding environment where the pack is located.
2.3	C-Rate	It refers to the ratio of the charge/discharge current to the pack capacity. For example, if the pack capacity is 314Ah and the charge/discharge current is 157A, the charge/discharge rate is 0.5C. Even if the pack capacity is reduced to 100Ah and the charge/discharge current is 50A, the charge/discharge rate remains 0.5C.
2.4	State of Charge (SOC)	It is the ratio of the pack capacity status to the nominal capacity under no-load conditions. It is measured in Wh and indicates the charge status of the pack. For example, if a capacity of 16,076Wh is 100% SOC, then 0Ah corresponds to 0% SOC at a current of 0.5C.
2.5	Open circuit voltage (OCV)	Open Circuit Voltage: The pack voltage measured under no-load conditions or when the circuit is disconnected. It is abbreviated as OCV.
2.6	Customer	Refers to the purchaser of the product sales contract concluded with Rimplier. Inc.

PART 3.
Product Features

Based on lithium iron phosphate batteries, it offers a long lifespan, safety, and reliability. The compact structural design effectively enhances system energy density. The standardized product design ensures excellent scalability. The BMS provides overcharge protection, over-discharge protection, discharge overcurrent protection, short-circuit protection, and overheating protection.

PART 4.

Product Dimensions & Specifications

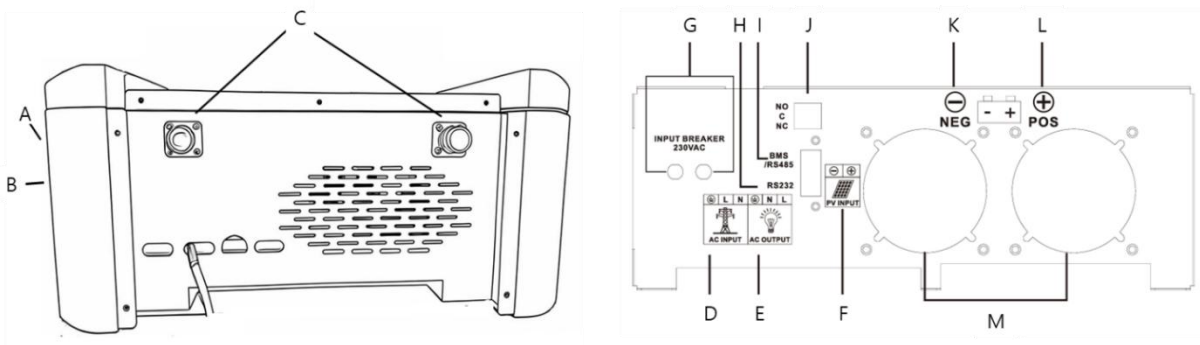
4.1 Battery pack



[Figure 1 : 16kWh battery pack exterior]

Item	Name	Description
A	(P+ Terminal)	400A
B	(P- Terminal)	400A
C	DC Circuit breaker	Circuit breaker (200A)
D	(Reset Button)	(Reset the system)
E	(LED Indicator)	(States, alarm, SOC)
F	(IP Switch)	Setting a unique IP address for parallel connection (Max. 16 address)
G	CAN / RS232 / RS485	Communication interface / RJ45 port
H	(ON/OFF Switch)	System ON/OFF
I	(Screen)	(4.3-Inch Touch screen)

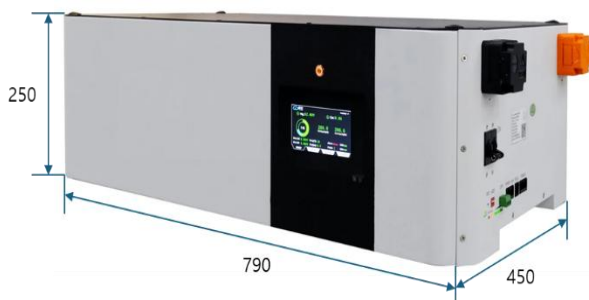
4.2 Inverter



[Figure 2 : 8kW Inverter Appearance]

Item	Name	Description
A	ON/OFF Switch	System ON/OFF
B	LCD screen	LCD screen / LED Indicator / Function keys
C	Battery input	51.2 V DC
D	AC input	220VAC / 110VAC
E	AC output	220VAC / 110VAC
F	PV input	
G	Input breaker	230VAC
H	RS232 communication port	RJ45 port
I	BMS/RS485 communication port	RJ45 port
J	Dry contact	3A / 250VAC
K	Battery terminal (-)	
L	Battery terminal (+)	
M	Fan	

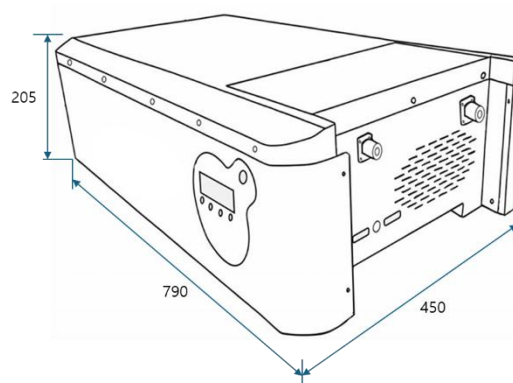
4.3 Product size



[Figure 3 : 16kWh battery pack size]

450(W) x 790(L) x 250(H) mm

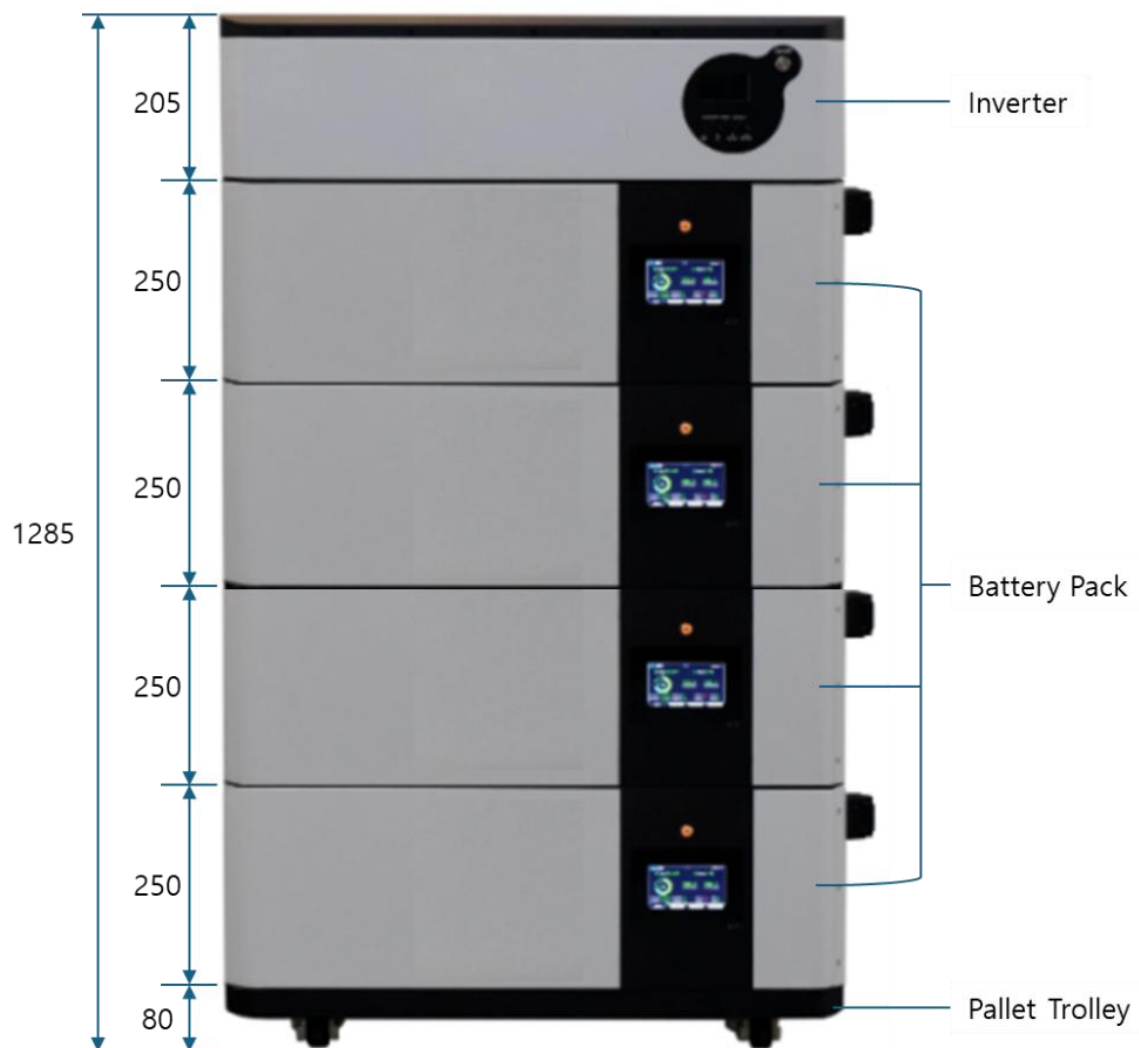
(Based on drawing; actual product may differ)



[Figure 4 : 8kW Inverter Size]

450(W) x 790(L) x 205(H) mm

4.4 Stack size



[Figure 5 : Battery pack 4*parallel and inverter stacking]

450(W) x 790(L) x 1285(H) mm

(Based on drawing; actual product may differ)

PART 5.
Product Specifications
5.1 Product Specifications - Battery Pack

No.	item	Specifications		Remarks
1	Nominal voltage	51.2 V		Standard charging method Standard discharging method
2	Nominal capacity	314 Ah		
3	Nominal energy	16.076 kWh		
4	Battery cell	Lithium iron phosphate battery (3.2 V 314 Ah)		LiFePO4
5	Module configuration	1P 16S		
6	Operation voltage range	40 V ~ 58.4 V		Cell voltage range: 2.5 V ~ 3.65 V
7	Standard charge current	157 A (0.5 C)		25 ± 2°C
8	Standard discharge current	157 A (0.5 C)		
9	Max. continuous discharge current	200 A (0.5 C)		
10	Dimensions	450mm x 790mm x 250mm (Width x Length x Height)		Size reference subject to physical objects
11	Weight	≤120Kg		Cells included
12	IP Rating	IP20		
13	Max. capacity expansion	16* Battery pack (parallel)		4* parallel: 64.307 kWh 16* parallel: 257.216 kWh
14	Max. vertical stacking	4* Battery pack + 1* Inverter (vertical)		Mobile trolley load limit : ≤500kg
15	Communication interface	CAN & RS485: > inverter communication RS232: parallel connection		RJ45 Port
16	Circuit protection	Overcharge protection / Overdischarge protection / Overcurrent protection High/low temperature protection / Short-circuit protection		
17	Safety equipment	250A Class-T Fuse, 200A Circuit Breaker,		
18	Fire protection	Automatic Fire Extinguisher System		
19	Operation environment	Temp. Charge	-10°C ~ 50°C	Ambient temperature
		Temp. Discharge	-20°C ~ 50°C	
		Humidity	RH 10% ~ 85%	
20	Storage environment	Humidity	RH 35% ~ 85%	
		Transportation and storage temperatures	-20°C ~ 55°C	SOC 40% ~ 60%

5.2 Product Specifications – Inverter

No.	item	Specifications	Remarks
Input			
1	Capacity	10,000 VA	
2	Voltage	48 V	DC
3	Nominal voltage	220V / 110V	AC
4	Voltage range	154-264V / 77-132V	AC
5	Frequency	50/60Hz	
6	AC Input Wiring	L+N+PE	
Output			
7	Rated power	8,000 W	
8	Voltage	220V / 110V	AC
9	Frequency	50 / 60Hz	
10	Waveform	sinusoidal waveform (Pure sinewave)	
11	Transfer time	< 8ms	AC ↔ DC
12	Output voltage regulation	10%rms	
13	Efficiency	> 98%	
14	Battery charging current	50A	
15	AC output Wiring	L+N+PE	
Photovoltaic Grid System			
16	Max. PV array power	6,400W	
17	MPPT/PWM input voltage range	MPPT 60-150V / PWM 60-105V	DC
18	Max. PV array open circuit voltage	MPPT 150V / PWM 105V	DC
19	Max. PV charging current	120A	
System			
20	Dimensions	450mm(W) x 790mm(L) x 205mm(H)	
21	Weight	About 30Kg	
22	Battery connection	Parallel : Min : 4 packs / Max : 16 packs	48 VDC
23	Communication interface	BMS & RS485: Battery pack communication RS232: Parallel connection	RJ45 Port
24	Operation environment	Operation Temp. : -10°C ~ 50°C	
		Humidity : RH 35% ~ 85%	
25	Storage environment	Temperature : -20°C ~ 55°C	
		Humidity : RH 35% ~ 85%	
26	Noise	< 45dB	

PART 6.**Instructions for use and precautions****6.1 Battery Pack Instructions**

Before using the product, carefully read the user manual and the markings on the product surface. Charging: Connect the battery pack's charging terminals to a compatible charger to charge (charging voltage is $62.05 \pm 0.5V$; do not reverse the polarity). Discharging: Connect the positive and negative terminals (do not reverse the polarity) indicated on the battery case to the corresponding load.

6.2 Precautions for Use

Keep the product away from heat sources and high voltages during use, and keep out of reach of children. Never short-circuit the positive and negative terminals. Do not disassemble the product arbitrarily, and do not allow it to get wet. This can be dangerous during use. If the product case is severely damaged or emits an unusual odor upon initial use, discontinue use and return it to the store. If the product will not be used for an extended period of time, charge the battery to 30% of its remaining capacity. Storing the battery in a discharged state for an extended period of time can cause overdischarge, shortening the product's lifespan or damaging the product.

PART 7.**Transportation and storage****7.1 Transportation**

All products are safely transported to the installation site in accordance with our battery/battery pack transportation specifications, using pallets or wooden packaging. During transportation, avoid exposure to severe vibration, shock, direct sunlight, or rain. Battery packs should be kept partially charged. The product should not be stored outdoors in direct sunlight or rain. Do not use sharp objects or forklifts to pierce the wooden packaging box. Wrap the exterior with stretch film to prevent moisture. Stack products no more than two layers high to prevent damage to the bottom product.

7.2 Storage

This product should be stored in a dry, clean, and well-ventilated area free from acidic, corrosive, or explosive materials at a 30% State of Charge (SOC). Avoid direct sunlight and store at least 2 meters away from heat sources. Do not stand the product upside down or subject it to excessive pressure. If stored for more than three months, check the voltage every three months to maintain the SOC of the battery module at 30-40%. If the SOC is too low, charge it.

PART 8.**Danger Warnings****8. Danger Warnings**

- (1) ****Do not disassemble or install this product or its built-in battery.**** This product has internal protection devices and circuitry to prevent accidents. Improper disassembly or installation may impair these protection functions, causing the battery to overheat, emit smoke, become deformed, or catch fire.
- (2) ****Do not short-circuit.**** Do not connect the positive and negative terminals of the product with metal objects. Also, do not store or move the product with metal objects. A short-circuit can cause excessive current to flow through the system, damaging the battery and resulting in overheating, smoke, deformation, or fire.
- (3) ****Do not heat or burn the product.**** Heating or burning the battery may melt the battery separator, lose its safety function, or burn the electrolyte. Overheating may cause the battery to overheat, emit smoke, become deformed, or catch fire.
- (4) ****Do not expose the product to rain or immerse it in water.**** This may cause the internal protection circuit to malfunction, resulting in an abnormal chemical reaction, which may cause the battery to overheat, emit smoke, become deformed, or catch fire. (5) ****Do not damage the product or battery.**** Do not damage the product by scraping the battery with metal objects, hitting it with a hammer, or dropping the product or battery. These actions may cause the battery to overheat, generate smoke, deform, or catch fire.
- (6) Do not connect the product directly to a power outlet. High voltage and current may pass through the battery, causing damage, overheating, smoke, deform, or catch fire.
- (7) Do not reverse charge. Reverse charging may damage the battery and cause overheating, smoke, deform, or catch fire.

PART 9.**Agreements**

Any other matters not included in this specification must be agreed between the manufacturer (Remplir Inc.) and the customer.

- (1) Any applications or operating conditions that differ from those specified in this document **must be discussed and approved** in advance by the manufacturer (Remplir Inc.). Additional testing may be required to verify the performance and safety of the product under such conditions.
- (2) The manufacturer shall not be held responsible for any accidents, damages, or malfunctions resulting from the use of the battery under conditions **not specified in, or not approved in writing by**, this document.
- (3) Any matters not covered in this specification shall be determined **through mutual agreement** between the manufacturer and the customer.
- (4) This product complies with the <Eco-Friendly and Hazardous Substance Management Standards for Manufactured Products>.

- (5) Manufacturer **reserves the right** to revise product specifications. The customer **must verify the latest version** of the product specification prior to purchase.
- (6) In the event of any discrepancy or inconsistency between the English and Korean versions of this document, **the Korean version shall prevail**.

PART 10.**Manufacturer Information**

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